

AUTOMOTIVE ENGINEERING

High-Level Design Document

Vehicle Climate Control ECU (VCC-ECU)

Document ID	VCC-HLD-001
Revision	1.0
Status	Engineering Draft
System	Vehicle Climate Control
ECU	Climate Control ECU
Architecture	AUTOSAR-style software architecture
Prepared For	Architecture Analysis Pilot

1. System Overview

The Vehicle Climate Control ECU (VCC-ECU) manages cabin temperature and blower operation. The software architecture is organized into application-level software components that communicate through typed ports and interfaces. The design uses an AUTOSAR-style component interaction model.

1.1 Primary Functions

The ECU shall acquire cabin temperature and ambient temperature, determine a target climate state, control blower speed, communicate vehicle state information, and provide diagnostic information.

1.2 Architectural Components

Component	Responsibility	Main Inputs	Main Outputs
SensorManager	Acquire and validate temperature sensor values	CabinTempRaw, AmbientTempRaw	CabinTemperature, AmbientTemperature
TemperatureManager	Calculate filtered temperatures and target climate	CabinTemperature, TargetTemperature	TemperatureDeviation
ClimateController	Determine climate operating mode and blower request	TemperatureDeviation, VehicleState	BlowerRequest, ClimateMode
FanController	Convert blower request into actuator command	BlowerRequest	BlowerSpeedCommand
VehicleStateManager	Provide vehicle operating state	IgnitionState, VehicleSpeed	VehicleState
DiagnosticManager	Monitor faults and expose diagnostic status	SensorStatus, ControllerStatus	DiagnosticStatus
CommunicationManager	Exchange climate signals with other ECUs	VehicleState, ClimateStatus	ClimateStatus

2. Interfaces and Ports

Software components communicate through interfaces. Sender-receiver communication is used for periodic vehicle and climate signals. The following interface catalogue is the authoritative interface inventory for this sample document.

Interface	Type	Provider	Consumer	Signals
ISensorData	Sender-Receiver	SensorManager	TemperatureManager	CabinTemperature, AmbientTemperature
ITemperatureControl	Sender-Receiver	TemperatureManager	ClimateController	TemperatureDeviation
IVehicleState	Sender-Receiver	VehicleStateManager	ClimateController	VehicleState
IBlowerControl	Sender-Receiver	ClimateController	FanController	BlowerRequest
IDiagnosticStatus	Sender-Receiver	DiagnosticManager	CommunicationManager	DiagnosticStatus
IClimateStatus	Sender-Receiver	ClimateController	CommunicationManager	ClimateStatus

2.1 Port Catalogue

Component	Port	Direction	Interface
SensorManager	P_CabinTemperature	P	ISensorData
TemperatureManager	R_CabinTemperature	R	ISensorData
TemperatureManager	P_TemperatureDeviation	P	ITemperatureControl
ClimateController	R_TemperatureDeviation	R	ITemperatureControl
ClimateController	R_VehicleState	R	IVehicleState
ClimateController	P_BlowerRequest	P	IBlowerControl
FanController	R_BlowerRequest	R	IBlowerControl
DiagnosticManager	P_DiagnosticStatus	P	IDiagnosticStatus
CommunicationManager	R_DiagnosticStatus	R	IDiagnosticStatus

3. Signals

Signal	Data Type	Range / Values	Source	Destination
CabinTemperature	float32	-40 to 85 degC	SensorManager	TemperatureManager
AmbientTemperature	float32	-40 to 85 degC	SensorManager	TemperatureManager
TargetTemperature	float32	16 to 30 degC	ClimateController	TemperatureManager
TemperatureDeviation	float32	-20 to 20 degC	TemperatureManager	ClimateController
VehicleState	uint8	0=OFF, 1=ACC, 2=RUN	VehicleStateManager	ClimateController
BlowerRequest	uint8	0 to 100 percent	ClimateController	FanController
BlowerSpeedCommand	uint8	0 to 100 percent	FanController	Actuator
DiagnosticStatus	uint16	0=OK; nonzero=fault	DiagnosticManager	CommunicationManager
ClimateStatus	uint8	0=OFF, 1=COOL, 2=HEAT, 3=AC OFF	ClimateController	CommunicationManager

4. Functional Flow

1. SensorManager reads raw cabin and ambient temperature values. 2. SensorManager validates the values and publishes temperature signals. 3. TemperatureManager filters the values and calculates TemperatureDeviation. 4. VehicleStateManager publishes VehicleState. 5. ClimateController evaluates TemperatureDeviation, VehicleState, and TargetTemperature. 6. ClimateController publishes BlowerRequest and ClimateStatus. 7. FanController converts BlowerRequest into BlowerSpeedCommand. 8. DiagnosticManager monitors sensor and controller status. 9. CommunicationManager publishes relevant climate and diagnostic status to other vehicle systems.

4.1 Dependency Summary

Upstream	Relationship	Downstream
SensorManager	provides ISensorData	TemperatureManager
TemperatureManager	provides ITemperatureControl	ClimateController
VehicleStateManager	provides IVehicleState	ClimateController
ClimateController	provides IBlowerControl	FanController
DiagnosticManager	provides IDiagnosticStatus	CommunicationManager
ClimateController	provides IClimateStatus	CommunicationManager

5. Architecture Notes

The ClimateController is the central application component for climate decision logic. TemperatureManager is responsible for temperature-derived information and does not directly control the blower. FanController is the only component responsible for translating BlowerRequest into an actuator command. DiagnosticManager is logically separated from the normal climate-control decision path.

6. Revision History

Revision	Date	Author	Description
0.1	2026-09-15	Architecture Team	Initial architecture draft
0.9	2026-09-25	Architecture Team	Added interface and signal catalogues
1.0	2026-10-01	Architecture Team	Pilot baseline for architecture analysis

Note: This is a synthetic engineering document created specifically for the AUTOSAR HLD AI Analyzer pilot. It is not an official AUTOSAR specification and does not represent a production vehicle architecture.